Bandung Institute of Technology

Informatics / School of Electrical Engineering and Informatics

IF5251 - AI in Production

Electricity Demand Forecasting

Group 04

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Chapter 1

Requirements

Electricity demand forecasting is a critical component of modern energy grid management. Accurate demand forecasts allow grid operators to balance supply and load proactively, reduce reliance on expensive peaking generation capacity, and facilitate the integration of variable renewable energy sources. This chapter defines the requirements for an AI-based electricity demand forecasting system that operates across multiple geographic locations, covering its goals, the needs of its intended users, the assumptions about the operating environment, and the metrics used to assess both model and system performance.

1.1 System and Model Goals

The *system goal* is to provide **accurate, automated, and near-real-time electricity demand forecasts** for multiple geographic locations, enabling energy operators and planners to make informed grid management decisions. The system ingests historical hourly consumption data alongside meteorological and temporal inputs spanning from 2016 onwards, and exposes forecast results through a web-based interface.

The *model goal* is to learn a **regression function** that maps a rich feature set which consists of over 30 meteorological variables. These features span from weather features such as temperature, humidity, solar radiation, and wind speed, to location identifiers and temporal features. The target variable, y, represents hourly electricity consumption per location cluster. Concretely, the model is expected to perform the following tasks.

- **Forecast** hourly electricity consumption for up to 24 hours ahead for each monitored location.
- **Generalise** across climatically and geographically diverse locations (e.g. London UK, Oakland CA, Groningen NL, Ottawa CA, Washington DC as the locations currently available).
- **Capture** non-linear relationships between weather drivers and demand, including seasonal patterns, diurnal cycles, and weather extremes.
- **Remain performant** on unseen future data through periodic retraining as new consumption records accumulate.

1.2 User Requirements

Based on domain analysis, the following user requirements have been identified in Table 1.1. These requirements are made on the basis of the intended use cases for the forecasting system, which include operational grid management, short-term energy procurement, and demand response planning.

In addition to that, these requirements assume that the users have a certain level of domain expertise in energy management and are familiar with interpreting time-series forecasts in the context of grid operations.

1.2.1 User Types

The system targets two primary user types: **energy grid operators**, who rely on short-term forecasts for real-time grid balancing decisions, and **energy planning analysts**, who use forecasts to support procurement and demand response planning. Both groups require domain expertise in energy management and are assumed to be comfortable interpreting time-series forecasts in an operational context.

The system is **not intended** for end consumers or non-technical stakeholders. Table 1.2 maps each user requirement to the applicable user type, along with representative usage scenarios and implementation priority.

1.2.2 Use Cases

Based on the user requirements defined earlier, the following use cases have been identified for the electricity demand forecasting system, as illustrated in Figure 1.1.

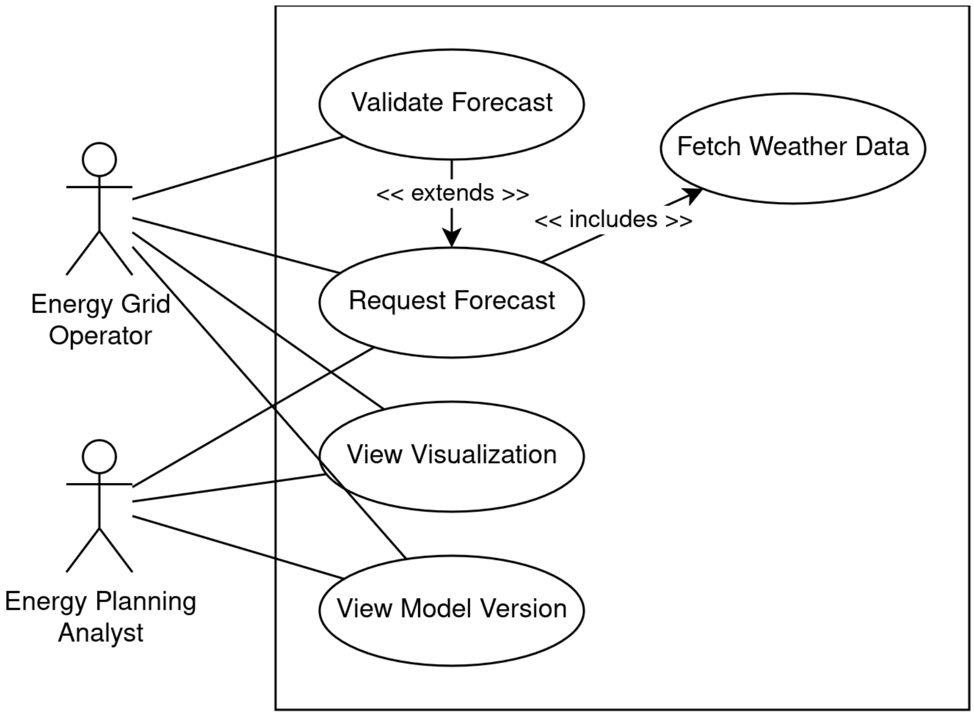

Figure 1.1: System Use Case Diagram

Each use case corresponds to a specific interaction between the user and the system, aligned with the requirements outlined in Table 1.1, Table 1.3 explains this further.

Table 1.3: Use Cases Overview

ID	Name	Description	Input	Output	Related Req.
UC-01	Request Forecast	User selects a location and time horizon to request a demand forecast via the web interface.	Location identifier, time horizon (up to 48 h)	Inference forecast result	UR-01, UR-02, UR-03, UR-07, UR-09
UC-02	View Visualisation	User views forecast results as a labelled hourly time-series plot.	Inference forecast result	Time-series plot rendered in the interface	UR-04
UC-03	Validate Forecast	User flags a low-confidence forecast for manual review. Extends UC-01.	Forecast result	Correction queued for retraining	UR-05, UR-08
UC-04	View Model Version	User checks which model version produced a given forecast.	Forecast identifier	Model version and training date	UR-06
UC-05	Fetch Weather Data	System automatically retrieves meteorological data for the requested location. Included in UC-01.	Location identifier, request timestamp	Variable meteorological feature set	UR-03

1.3 Environment Assumptions

Environment assumptions explicitly define the conditions under which the system is expected to operate correctly. They bound the system's scope and serve as the basis for design decisions made in subsequent chapters. The assumptions are listed in Table 1.4.

Deviations from these assumptions may require system or model updates, and any new assumptions that arise during development should trigger a reassessment of the relevant requirements.

1.4 System Components and Interactions

This section identifies the AI and non-AI components that constitute the forecasting system, maps their interactions, and describes the feedback loops through which the system improves over time.

1.4.1 AI and Non-AI Components

Table 1.5 categorises the system components into AI and non-AI elements, along with their roles and interactions.

1.4.2 Feedback Loop

The system incorporates an **operator correction loop** that enables continuous model improvement. Figure 1.2 illustrates this feedback mechanism.

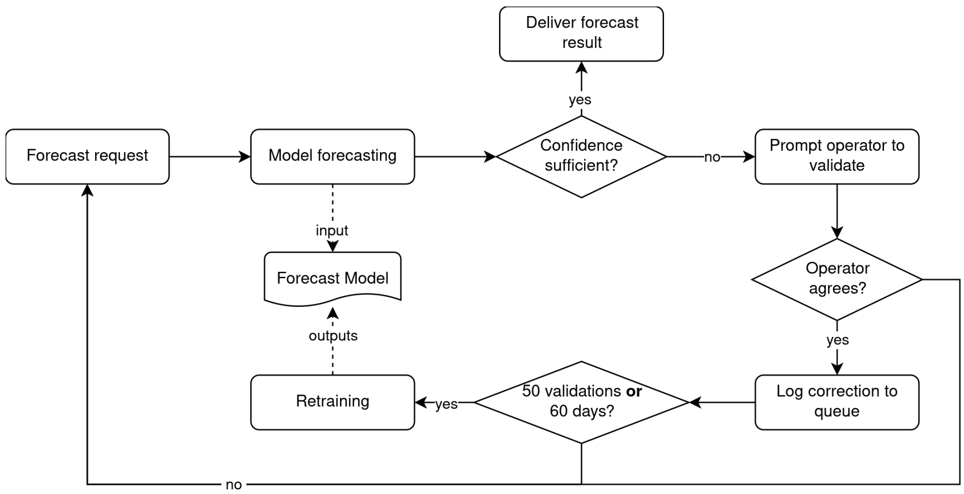

Figure 1.2: System Feedback Loop

When a forecast request is received, the model produces a result which is then checked against a confidence threshold. If confidence is sufficient, the forecast is delivered directly to the operator. Otherwise, the operator is prompted to validate the forecast.

If the operator agrees, the correction is logged to the validation queue; if not, the forecast is dismissed without further action. Once the queue reaches 50 validated corrections, or 60 days have elapsed since the last retraining, whichever comes first, a retraining job is triggered. The retrained model then replaces the previous one and resumes serving forecast requests.

1.5 Metrics for Measurement

Measuring system performance requires a combination of model-level (functional) and system-level (non-functional) metrics. Functional metrics assess the accuracy and reliability of the demand forecasts produced by the regression model. Non-functional metrics assess the operational characteristics of the deployed service, including responsiveness, scalability, and availability.

1.5.1 Functional Metrics

The following functional metrics are used to evaluate the regression model's forecasting accuracy. All metrics are computed on a held-out test set comprising data from a time period not seen during training, across all monitored locations. For each metric, specific target thresholds are defined based on domain standards and operational requirements for electricity demand forecasting, which are shown in Table 1.6.

- **MAE** (Mean Absolute Error): the average absolute deviation between predicted and actual consumption; interpretable in the same unit as the target variable and robust to outliers.
- **RMSE** (Root Mean Squared Error): penalises large forecast errors more heavily than MAE, making it sensitive to extreme deviations that are operationally costly.
- **MAPE** (Mean Absolute Percentage Error): expresses error as a percentage of actual consumption, enabling fair comparison across locations with different absolute demand scales.
- $\mathbf{R^2}$ (Coefficient of Determination): measures the proportion of variance in the target variable explained by the model, where a value of 1.0 indicates a perfect fit.

1.5.2 Non-Functional Metrics

Non-functional metrics govern the quality of the deployed service independent of model accuracy. These metrics ensure the system is operationally reliable for production use by energy grid operators operating on a 24/7 basis. For each non-functional metric, specific target thresholds are defined based on the operational requirements of real-time forecasting services in the energy sector, which are shown in Table 1.7.

- **Latency**: the end-to-end response time from API request submission to forecast delivery. Low latency is critical in grid management contexts where decisions are time-sensitive.
- **Throughput**: the number of concurrent forecast requests the service can handle per unit time without degradation, ensuring the system scales to multiple simultaneous operator queries.
- **Availability**: the percentage of time the forecasting service is accessible. Unplanned downtime has direct operational consequences for grid operators who depend on continuous access.
- **Model Freshness**: the maximum permissible age of the deployed model before retraining is mandated, ensuring that concept drift and seasonal pattern shifts do not degrade forecast accuracy over time.

Table 1.7: Non-Functional Metrics Overview

Metric	**Target**
Latency	$< 500\,\text{ms}$
Throughput	≥ 50 req/s
Availability	$\geq 99.5\%$
Model Freshness	≤ 60 days

1.6 Failure Mode and Effects Analysis

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo.

Figure 1.3: FMEA Diagram Placeholder

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero.

Table 1.1: User Requirements Overview

ID	Requirement Name	Description	Rationale
UR-01	Forecast Query	Users shall be able to request an electricity demand forecast for a specific monitored location and a time horizon of up to 48 hours via a web-based interface.	Enables users to obtain actionable forecasts for operational decision-making.
UR-02	Location Selection	The system shall support all locations present in the training dataset, selectable by name or geographic identifier (e.g. city name or `location_id`).	Allows users to access forecasts relevant to their specific geographic area of interest.
UR-03	Weather Data Fetch	The system shall fetch the current relevant weather data for each location automatically.	Reduces user burden by eliminating manual weather input, while ensuring forecasts are based on current meteorological data.
UR-04	Forecast Visualisation	Forecast results shall be displayed as a time-series plot of predicted hourly consumption over the selected time window, with clear axis labelling.	Facilitates intuitive understanding of forecast trends and patterns.
UR-05	Validation Interface	Users must be able to flag or manually validate forecasts that fall below a confidence threshold.	Allows users to ensure forecast quality and accuracy and supplying additional data for retraining.
UR-06	Model Version Visibility	Users must be able to see which version of the forecasting model was used to generate a particular forecast.	Enables users to understand the context and potential staleness of each forecast.
UR-07	Response Timeliness	Forecast results shall be returned within an operationally acceptable response time to support real-time grid management decisions.	Ensures forecasts are delivered in a timeframe that allows for timely action.
UR-08	Accuracy Transparency	The interface shall communicate expected model error ranges so users can appropriately calibrate their trust in the forecasts.	Builds user trust by setting realistic expectations about forecast reliability.
UR-09	Continuous Availability	The forecasting service shall be accessible at all times, reflecting the 24/7 nature of grid operations.	Ensures users can access forecasts whenever needed without interruption.

Table 1.2: User Requirements Mapping to System Users

ID	Energy Grid Operator	Energy Planning Analyst	Usage Scenario	Priority
UR-01	✓	✓	Requesting forecasts for operational decision-making.	High
UR-02	✓	✓	Selecting specific locations for forecast retrieval.	High
UR-03	✓	✓	Ensuring queries doesn't burden the user.	High
UR-04	✓	✓	Visualising forecast results for interpretation.	Medium
UR-05	✓	✗	Validating forecasts that fall below confidence thresholds.	Medium
UR-06	✓	✓	Seeing which version of the model was used.	Medium.
UR-07	✓	✗	Ensuring forecasts are delivered in a timely manner for real-time actions.	High
UR-08	✓	✓	Understanding expected error ranges to calibrate trust.	Medium
UR-09	✓	✓	Accessing forecasts at any time without interruption.	High

Table 1.4: Environment Assumptions Overview

ID	**Assumption Name**	**Description**
EA-01	Weather Data Availability	Real-time and forecast meteorological data matching the 30+ feature schema used during training is continuously available from the Weather API as model input at inference time.
EA-02	Hourly Temporal Resolution	All historical and incoming data is timestamped at hourly granularity. Sub-hourly or irregular inputs are outside the current scope.
EA-03	Stationarity of Demand Patterns	The relationship between weather conditions, time features, and electricity consumption is assumed to be sufficiently stable over time for the trained model to remain valid between retraining cycles. Major structural disruptions may require earlier retraining.
EA-04	Multi-Location Heterogeneity	Each location has a distinct demand profile driven by local climate, building stock, and population density. The location_id and cluster_size fields capture this heterogeneity and must be present at inference time.
EA-05	Data Quality	Historical data is assumed to have been preprocessed (null values handled, outliers addressed) as reflected in the full_data_non_null.csv dataset. Incoming data at inference time is expected to pass basic schema and range validation checks before being fed to the model.
EA-06	Deployment Infrastructure	The system is deployed on a server or cloud instance capable of hosting the trained model and serving REST API requests from the Next.js frontend. Internal component communication is assumed to be stable enough to meet the 500 ms latency target.
EA-07	Weather API Reliability	The external weather data provider is assumed to maintain sufficient uptime and data consistency. Prolonged API unavailability or schema changes may require fallback handling or data pipeline updates.
EA-08	Location Scope	The system only serves locations present in the training dataset. Queries for out-of-distribution locations are outside the current scope and will not produce valid forecasts.
EA-09	Access Control	The system is assumed to be deployed in a controlled environment accessible only to authorized personnel. No public-facing authentication layer is required within the current scope.

Table 1.5: System Components Overview

Type	**Component**	**Role**
AI	Forecasting Model	Produces hourly electricity demand forecasts and confidence estimates.
Non-AI	Inference Service	Serves model inference requests and routes responses to the frontend.
Non-AI	Feature Store	Manages and serves consistent feature sets for both training and inference.
Non-AI	Database	Persists historical consumption data, forecast records, and validation logs.
Non-AI	Message Queue	Buffers operator validations and triggers retraining when the queue threshold is reached.
Non-AI	Dashboard Service	Exposes a read API for the frontend to query historical weather and electricity consumption records from the database.
Non-AI	Frontend Website	Provides the web interface for forecast queries, visualisation, and validation prompts.

Table 1.6: Functional Metrics Overview

Metric	**Description**	**Target**
MAE	Mean Absolute Error	$< 10\%$ of mean target per location
RMSE	Root Mean Squared Error	$< 15\%$ of mean target per location
MAPE	Mean Absolute Percentage Error	$< 10\%$
R^2	Coefficient of Determination	> 0.85

Chapter 2

Architecture and Design

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis.

2.1 Modelling Tradeoffs

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2.2 Deployment Architecture

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Figure 2.1: Deployment Architecture Diagram Placeholder

Aliquam lectus. Vivamus leo.

2.3 Data Science Pipeline

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim.

Figure 2.2: Data Science Pipeline Diagram Placeholder

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi.

2.4 Telemetry and Monitoring

Nulla mattis luctus nulla. Duis commodo velit at leo. Aliquam vulputate magna et leo. Nam vestibulum ullamcorper leo.

2.5 Anticipate Evolution

Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio.

2.6 Big Data Processing

Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh. Maecenas non sem quis tortor eleifend fermentum.

Figure 2.3: Big Data Processing Architecture Placeholder

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra.

2.7 Human-AI Design

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus.

Chapter 3

Quality Assurance

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor.

3.1 Model Testing

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci.

Table 3.1: Model Testing Results Placeholder

Test Case	**Expected Output**	**Actual Output**
Case 1	Placeholder	Placeholder
Case 2	Placeholder	Placeholder
Case 3	Placeholder	Placeholder

3.2 Data Quality

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis.

3.3 QA Automation

Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem.

3.4 Testing in Production

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum.

Adversarial Attacks

Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim.

Load Testing

Donec vel nibh ut felis consectetuer laoreet. Donec pede. Sed id quam id wisi laoreet suscipit.

3.5 Infrastructure Quality

Donec molestie, magna ut luctus ultrices, tellus arcu nonummy velit, sit amet pulvinar elit justo et mauris. In pede. Maecenas euismod elit eu erat. Aliquam augue wisi, facilisis congue, suscipit in, adipiscing et, ante.

3.6 Debugging

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Chapter 4

Operations

Sed mattis, erat sit amet gravida malesuada, elit augue egestas diam, tempus scelerisque nunc nisl vitae libero. Sed consequat feugiat massa. Nunc porta, eros in eleifend varius, erat leo rutrum dui, non convallis lectus orci ut nibh.

4.1 Unit Testing, Integration Testing and Continuous Training Pipeline

Sed consequat tellus et tortor. Ut tempor laoreet quam. Nullam id wisi a libero tristique semper. Nullam nisl massa, rutrum ut, egestas semper, mollis id, leo.

Figure 4.1: Continuous Training Pipeline Diagram Placeholder

Phasellus id magna. Duis malesuada interdum arcu.

4.2 Configuration Management

Sed eleifend, eros sit amet faucibus elementum, urna sapien consectetuer mauris, quis egestas leo justo non risus. Morbi non felis ac libero vulputate fringilla. Mauris libero eros, lacinia non, sodales quis, dapibus porttitor, pede. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos.

4.3 Monitoring

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Figure 4.2: Monitoring Dashboard Placeholder

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio.

4.4 Versioning

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst.

4.5 MLOps

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim.

Figure 4.3: MLOps Workflow Diagram Placeholder

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo.

Chapter 5

Responsible AI

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus.

5.1 Privacy

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis.

5.2 Sensitive Audits

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit.

Table 5.1: Sensitive Audit Checklist Placeholder

Audit Item	**Status**	**Notes**
Data Anonymisation	Pending	Placeholder
Bias Assessment	Pending	Placeholder
Fairness Evaluation	Pending	Placeholder

5.3 Explainability

Quisque consectetuer. In suscipit mauris a dolor pellentesque consectetuer. Mauris convallis neque non erat. In lacinia.

Figure 5.1: Model Explainability (SHAP / LIME) Placeholder

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu.

5.4 Transparency

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